Higher National Diploma in Engineering (CE)

04th Year, First Semester Examination - 2021

CE 3222 – Foundation Engineering

Instructions for Candidates:
Answer five questions only

No. of questions: 06
No. of pages: 07
Time: 03 hours

Q1).

a) State three soil properties that can be found from the Standard penetration test. (4 marks)

b) What are the advantages of the cone penetration test over the Standard penetration test? (state two advantages) (2 marks)

c) Briefly explain the Cone penetration test procedure? (6 marks)

d) As a part of the site investigation Standard Penetration Test (SPT) was conducted and it was observed that the field N-value was 15 for the depth of 6m from the ground level, the standard penetration test was performed with a standard split spoon sampler and hand dropped donut hammer with 60% energy efficiency. The bore diameter was 100mm. the length of the drilling rod that connects to the split spoon sampler is 10m. Assuming that there is no linear inside the sampler calculate the corrected SPT N value. (N_{70}) (Hint: Density of the soil layer $\gamma =$ 17KN/m3, 1Pa = 1N/m^2) (8 marks)

$$C_N = \sqrt{\frac{95.76}{\sigma'_v}}$$

Q2).

a) Using the phase diagram of soil element state void ratio, porosity, moisture content, unit weight, dry unit weight, and Saturated unit weight using the weight volume notifications.

(3 marks)

b) If the volume of the soil solids ($V_s = 1$), and Using the above definitions define the following. (Given: $G_s = \frac{W_s}{\gamma_w}$)

I. $(density)\gamma = \frac{(1+w)G_s \cdot \gamma_w}{1+e}$

II. $\gamma_{sat} = \frac{(G_s+e)\cdot\gamma_w}{1+e}$

(6 marks)

c) The saturated unit weight of a soil is 20.1 kN/m³, given that the Specific gravity (G_s = 2.74,), Assume the density of water γ_w = 9.81kN/ m³, determine the following values,

I. Dry density
II. Void ratio
III. Porosity
IV. Water content

(8marks)

d) State three types of pile testing methods. (3mrks)

Q3).

a) Write short notes on the following retaining structures describing the areas provided within the brackets.

I. Gabion walls (Materials used, stability of the structure, locations it used, purpose of use)
II. Diaphragm walls (Materials used, stability of the structure, locations used, purpose of use)

(6 marks)

b) The cross-section of the cantilever retaining wall is shown in Fig Q3.1. Calculate the factor of safety with respect to overturning and sliding. $\gamma_{concrete}$ = 24 kN/m³.
(Hint: $\sigma = \gamma z k_a - 2c\sqrt{k_a}$)

(14 marks)

Q4).

a) A circular type plie with 0.5m diameter embedded in to a sandy soil layer, of γ = 17.3 kN/m³ and φ = 30°. The length of the pile is 10 m. find the load carrying capacity of the pile. following equations are present with the notations with their usual meaning

$Q_u = Q_p + Q_s$

$Q_p = A_p q' N_q^* \leq A_p q_1$ and $q_1 = 50 N_q^* tan\varphi$

$Q_s = \sum p.\Delta L f$, $f = K.\sigma'_0 tan\delta$ and $L' = 15D$, K = 1.3 and $\delta = 0.8\varphi$ (10marks)

a) A square concrete pile 20m long with a cross section of 381mm × 381mm is fully embedded in a saturated clay layer. For the clay layer, γ_{sat} = 18.5 kN/m³, φ = 0, and $C_u = 70\ kN/m^2$. Assume that the water table lies below the tip of the pile. Determine the allowable load that the pile can carry (FS= 3). Use the αmethod to estimate the skin resistance

(10 marks)

Q5).

a) Derive an expression for Rankine active earth pressure developed in the granular soil using the Morh-Coulomb failure envelope. (5 marks)

Details of a cantilever retaining wall are shown in Fig Q5.1. A surcharge pressure of 10KN/m2 is being carried on the surface behind the wall. Due to the poor drainage, the water table has risen to the level indicated at the level shown in the Figure Q5.1. The unit weight and shear strength parameters of each soil layers are given in the figure. You may use the unit weight of water as 9.81 kN/m³

b) Plot the distribution of the effective lateral earth pressure diagram behind the retaining wall (without considering the pore water pressure).

(4marks)

c) Plot the lateral earth pressure due to pore water (presence of the water table) behind the retaining wall.

(4 marks)

d) Plot the resultant total lateral earth pressure diagram behind the retaining wall.

(2 marks)

e) Calculate total lateral force and determine it point of application.

(5marks)

Q6).

a) Briefly explain the importance of site investigation and list the information gathered from the site investigation. (5marks)

b) Determine the size of a square footing to carry net allowable load of 295KN, FS =3. Use Terzaghi's equation assuming general shear failure. (7marks)

c) Determine the net Ultimate bearing capacity of a mat foundation measuring 13m× 9m (L× B) has to be constructed on a saturated clay $C_u = 94\ KN/m^3$.

$\varphi = 0$ *and* $D_f = 2m$. (3marks)

Hint: $q_{(net)} = 5.14C_u\left(1 + \frac{0.195B}{L}\right)\left(1 + \frac{0.4D_f}{B}\right)$

d) State five reasons for pile failures and explain them.

(5 marks)

Table Q1: Correction factors

	Hammer for η_1				**Remarks**
	Donut		Safety		
Country	R-P	Trip	R-P	Trip/Auto	R-P = rope- pulley or cathead
United States					$\eta_1 = {E_r}/{E_{rb}}$
North America	45	-	70 - 80	70 - 100	For U.S trip/auto w/E_r = 80
Japan	67	78	-	-	$\eta_1 = \frac{80}{70} = 1.14$
United Kingdom	-	-	50	60	
China	50	60	-	-	
	Rod Length correction η_2				
Length	>10m		$\eta_2 = 1.00$		N is too high for L < 10m
	6-10m		$\eta_2 = 0.95$		
	4-6m		$\eta_2 = 0.85$		
	0-4m		$\eta_2 = 0.75$		
	Sampler correction η_3				
Without linear			$\eta_3 = 1.00$		Base value N is too high with linear
With linear : dense sand , clay			$\eta_3 = 0.80$		
With linear: loose sand			$\eta_3 = 0.90$		
	Borehole diameter correction η_4				
Hole diameter	60-100mm		$\eta_4 = 1.00$		Base value: N is too small when have an oversize hole
	150mm		$\eta_4 = 1.05$		
	200mm		$\eta_4 = 1.15$		